

Code Manual: Project 11

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PDF VERSION	Initial version
CODE VERSION	0.6.x
REPOSITORY LINK	https://github.com/AineMedias/CQM

Abstract

This program is designed to take three potentials and create H -matrices from them, using the method outlined in the 2013 paper by Jugdutt[1], in order to determine the energy eigenvalues arising from said potentials.

1 Mathematical model

2 Algorithmic overview

The centerpiece of this entire program revolves around the `eigenvalues()` function, which uses a predefined potential V and upper bound n_{max} to determine the two-dimensional Hamiltonian matrix H_{nm} according to the formula outlined in the paper by Jugdutt[1]:

Since the Python file has its constants expressed in atomic units, where $\hbar = 1$, $e = 1$, the electronic mass $m_0 = 1$ and the permeability Firstly, all necessary

3 Software interface and code execution

In order to effectively run this program, Python version 3.10 or later is recommended to be installed on the virtual environment that the program is supposed to run on. Below this version, parts of the Python file's main function that use the match-case system will not work and raise exceptions accordingly. Furthermore, assuming the virtual environment was freshly made to run this specific program, the matplotlib, numpy, scipy and tqdm packages must be installed first, either through `pip install` or, if the Anaconda distribution is currently in use, through `conda install` within a PowerShell instance.

If all required packages are installed, the code can simply be run through the command `python3 centralpotentials.py`, or `& envpath "filepath" in Anaconda`, where `envpath` represents the path leading to the virtual environment's `python.exe` file, and `filepath` represents the path leading to the downloaded `centralpotentials.py` file.

In both cases, before execution of the `eigenvalues` function or loop containing said function begins, the user is asked to insert a string representing the quantum system to be examined: the supported strings are `h-atom` representing the Coulomb potential, `fin-well` representing the finite spherical well and `yukawa` for the Yukawa potential. Any other strings raises a `ValueError`.

Next, the user is asked whether they want to examine the potential at a given value or execute the default function on these: they are supposed to type in respectively 1 or 0 before continuing on, otherwise another `ValueError` is raised. Dependent on the selected potential, more variables might be required as input, such as the screening constant μ for the Yukawa potential or the well depth V_0 and width b for the finite spherical well.

Afterwards, code execution begins - in case that a peculiar potential is to be examined or the selected potential is `h-atom`, the program will run `eigenvalues()` once, and generate a graph where the energy eigenvalues in Hartree units ($E_h = 27.211$ eV) are defined as a function of n . If the selected potential is `fin-well` or `yukawa` and the default function variable is 0, the program will instead loop over

respectively V_0 and b , or over V in Hartree units, and instead returns a list for `fin-well`, or the critical coefficient μ_c for `yukawa`.

4 Benchmark tests

5 Performance

Each run of

6 References

- [1] BA Jugdutt and F Marsiglio. "Solving for three-dimensional central potentials using numerical matrix methods". In: *American Journal of Physics* 81.5 (2013), pp. 343–350.